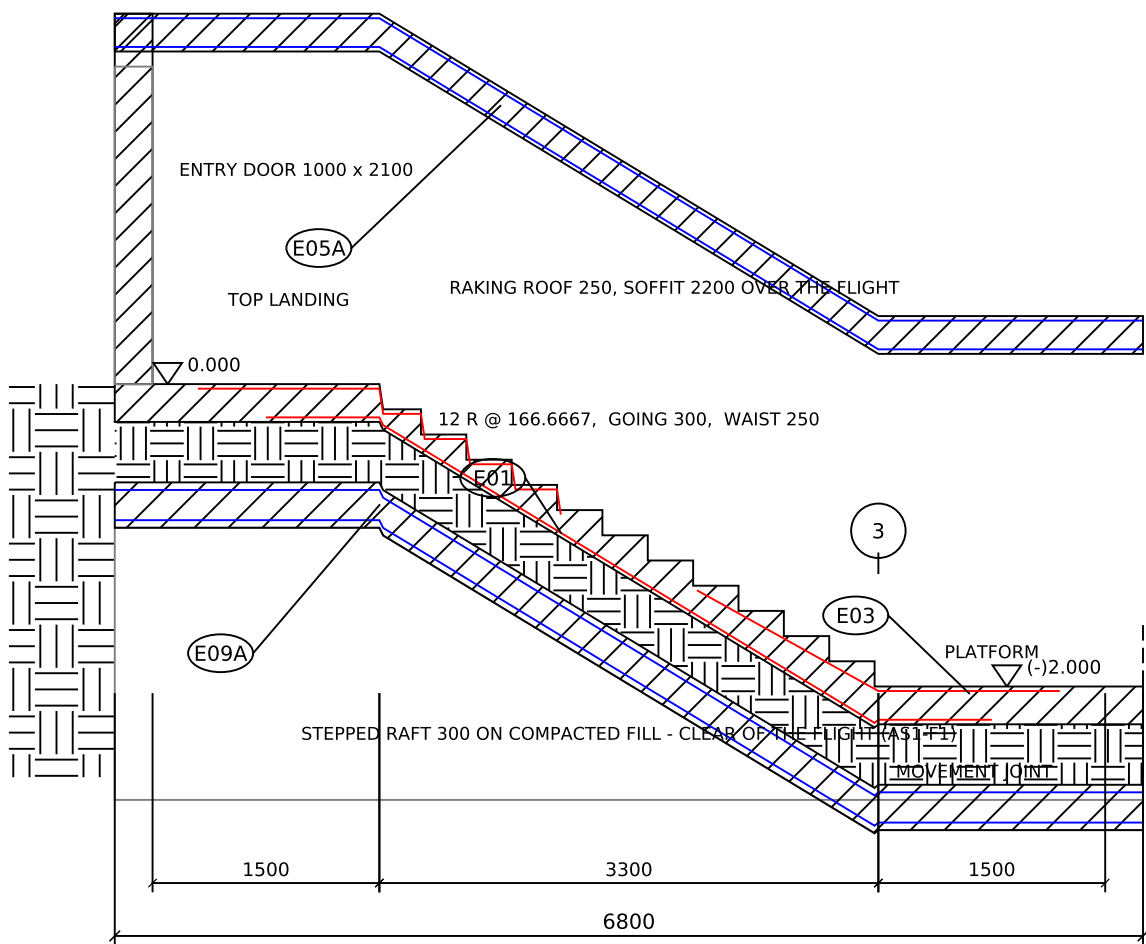
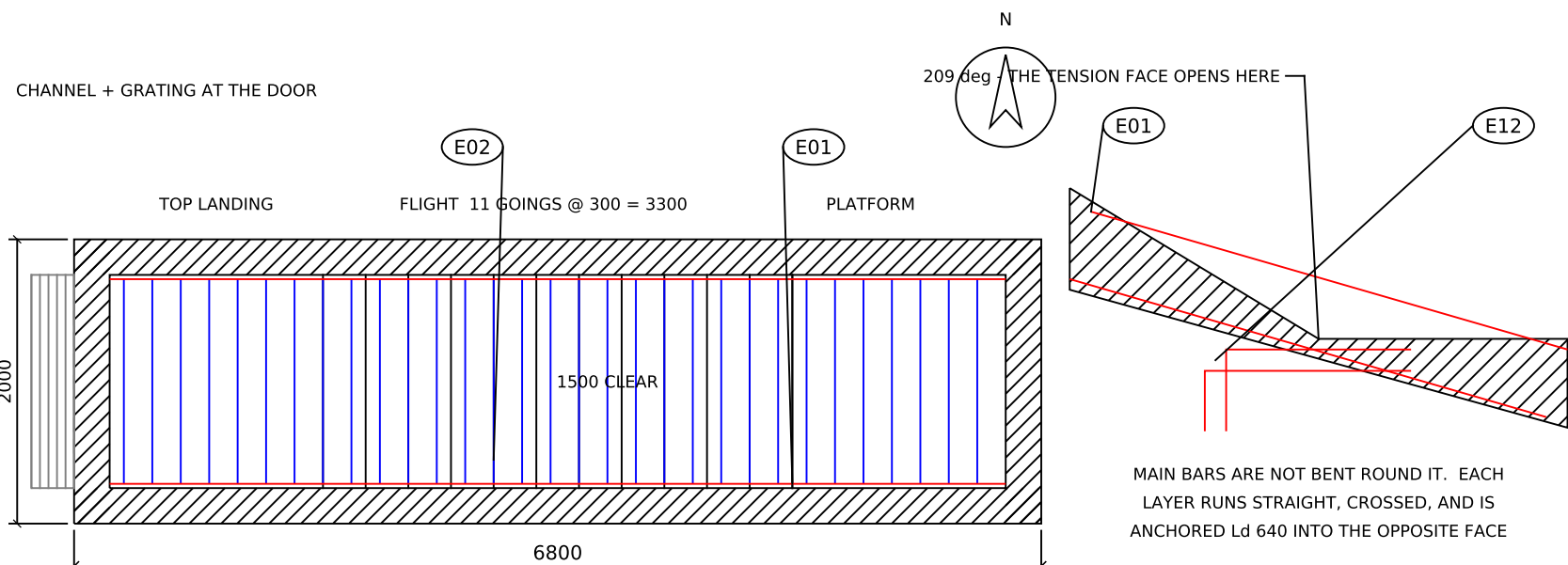


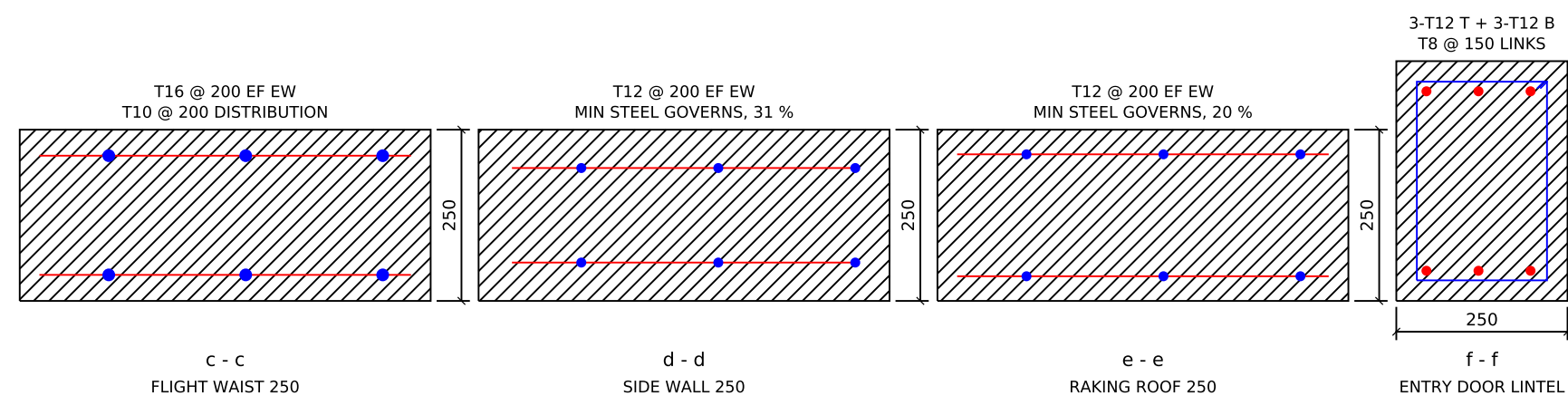


1 ENTRY STAIRWELL - LONGITUDINAL SECTION B-B
1 : 50 CUT ON THE 1500 WIDTH CENTRELINE, LOOKING NORTH



2 ENTRY STAIRWELL - PLAN
1 : 50 EXTERNAL 6800 x 2000, INTERNAL 6300 x 1500, WALLS 250

3 THE OPENING CORNER
1 : 20 TOP OF THE FLIGHT, SP 34 Cl. 5.5



4 SECTIONS - FLIGHT, WALL, ROOF AND LINTEL
1 : 10 COVER 30 INTERNAL, 50 EARTH FACE

ENTRY STAIRWELL - ELEMENT SCHEDULE						
ELEMENT	THK	COVER	d	MAIN	DISTRIB.	TOP STEEL
FLIGHT, WAIST 250	250	30	214	T16 @ 200 (1005)	T10 @ 200	T16 @ 200, 1200 INTO THE FLIGHT
TOP LANDING 0.000	250	30	214	T12 @ 200 EF EW (565)	-	-
PLATFORM (-)2.000	250	30	214	T12 @ 200 EF EW	-	ON FILL
SIDE WALLS	250	50	194	T12 @ 200 EF EW	-	MIN STEEL GOVERNS
RAKING ROOF	250	30	204	T12 @ 200 EF EW	-	MIN STEEL GOVERNS
HEADWALL	250	30	214	T12 @ 200 EF EW	-	-
DOOR LINTEL 1000 CLEAR	250 x 350	30	306	3-T12 T + 3-T12 B	T8 @ 150	-
STEPPED RAFT	300	50	244	T12 @ 200 EF EW	-	ON COMPACTED FILL
OUTSIDE THE PROTECTIVE BOUNDARY. NOT BLAST RATED. DESIGNED TO IS 456, NOT TO IS 4991						

BAR MARK SCHEDULE - COVERED ENTRY STAIRWELL					
MARK	BAR	SPACING	LOCATION	CUT mm	No.
E01	T16	200	MAIN BARS	5275	8
E02	T10	200	DISTRIBUTION	1440	19
E03	T16	200	TOP, 1200 INTO THE FLIGHT	1840	16
E04A	T12	200	VERTICAL, EACH FACE	3500	136
E04B	T12	200	HORIZONTAL, EACH FACE	6700	60
E05A	T12	200	TRANSVERSE, EACH FACE	1900	68
E05B	T12	200	LONGITUDINAL, EACH FACE	6700	20
E06	T12	200	BOTH WAYS, EACH FACE	1900	32
E07	T12	200	BOTH WAYS, EACH FACE	1900	32
E08A	T12	200	VERTICAL, EACH FACE	3500	20
E08B	T12	200	HORIZONTAL, EACH FACE	1900	30
E09A	T12	200	LONGITUDINAL, EACH FACE	6700	20
E09B	T12	200	TRANSVERSE, EACH FACE	1900	66
E10	T12	-	3-T12 TOP + 3-T12 BOTTOM	1960	6
E11	T8	150	LINKS @ 150	1120	13
E12	T16	200	U-BAR ACROSS THE CORNER	1780	8

ENTRY STAIRWELL - DESIGN SUMMARY						
ELEMENT	LOAD	SPAN / BASIS	M	Ast,req	PROVIDED	UTIL
FLIGHT, WAIST 250	wu 22.85 kPa	Leff 4800 (Cl. 33.1(b))	M 65.8 kNm/m	Ast,req 744	T16 @ 200 (1005)	75 %
SIDE WALLS 250	34 kPa AT BASE	RETAINED 2.9 m, PROPPED	M 14.0 kNm/m	168 ; min 500	T12 @ 200 EF EW (565)	31 %
RAKING ROOF 250	wu 50.5 kPa	1500 CLEAR	M 9.5 kNm/m	108 ; min 300	T12 @ 200 EF EW	20 %
TOP LANDING 250	wu 54.9 kPa	Leff 1750	M 21.0 kNm/m	Ast,req 229	T12 @ 200 EF EW (565)	41 %
THE RAKING ROOF CARRIES A 20 kPa IMPOSED LOAD, DELIBERATELY, FOR A STRAY VEHICLE ON THE BERM						
FLIGHT DEFLECTION $4800/214 = 22.4$; basic 20 ; fs 215 ; pt 0.470 % ; MF about 1.5 -> 30. SHEAR $\tau_{au_v} 0.256 < \tau_{au_c} 0.484 \times k 1.10$ - NO LINKS						

CONSTRUCTION REQUIREMENTS THAT ARE DESIGN OUTPUTS, NOT OPTIONS	
1 THIS STRUCTURE IS OUTSIDE THE PROTECTIVE BOUNDARY AND IS NOT BLAST RATED. It is expected to be LOST in the design event. Designed to IS 456 with normal partial factors - NOT with IS 4991 dynamic strengths. The blast boundary is the headhouse wall HW2 and its door.	
2 THE OPENING CORNER AT THE TOP OF THE FLIGHT IS THE DETAIL MOST OFTEN GOT WRONG. There the tension face turns through 209 deg, and a bar bent round that corner has its bend resultant directed OUT of the concrete: it spalls the cover and the bar loses its anchorage. *** MAIN BARS SHALL NOT BE BENT ROUND IT. *** Each layer runs STRAIGHT, CROSSED, anchored Ld = 640 into the OPPOSITE face, with a U-bar T16 @ 200 (mark E12) across the corner. SP 34:1987 Cl. 5.5. At the FOOT of the flight the corner CLOSES at 151 deg - bars turn normally there.	
3 MOVEMENT JOINT WHERE THE STEPPED RAFT MEETS THE HEADHOUSE. The raft bears on compacted fill, the headhouse on the pressure slab. Different supports, different settlement - they must NOT be cast monolithic.	
4 300 mm CHANNEL AND GRATING THE FULL 1500 WIDTH at the door (X 8950 - 9250), and a 1.0 m3 external sump at the platform with its own soakaway. The stairwell has no gravity outfall to the shelter drainage and must not be allowed to discharge into it.	
5 BERM GRADED 1.5 : 1 AGAINST THE WALLS TO (+)0.900, toe at grade over 1350. The 20 kPa roof imposed load is what covers a stray vehicle on that berm.	

UNDERGROUND CBRN-HARDENED
BLAST-RESISTANT PROTECTIVE
STRUCTURE + SENTRY POST
PUNE, MAHARASHTRA

NOTES

- ALL DIMENSIONS ARE IN MM. ALL LEVELS ARE IN METRES RELATIVE TO FINISHED SITE GRADE 0.000, NEGATIVE DOWNWARDS. WRITTEN DIM TO BE FOLLOWED ONLY. DRG NOT TO BE SCALED.
- THIS DRG IS TO BE READ IN CONJUNCTION WITH ALL RELEVANT ARCH AND STR DRGS. CONCRETE M35, w/c NOT MORE THAN 0.45. REINFORCEMENT Fe500D TO IS 1786:2008. Ld 40 PHI, LAPS 50 PHI, ALL STAGGERED SO THAT NOT MORE THAN 50 % ARE SPLICED AT ONE SECTION.
- *** THIS STRUCTURE IS OUTSIDE THE PROTECTIVE BOUNDARY AND IS NOT BLAST RATED (REV F DRAWING NOTE 8). IT IS EXPECTED TO BE LOST IN THE DESIGN EVENT. IT IS DESIGNED TO IS 456 WITH NORMAL PARTIAL FACTORS, NOT WITH IS 4991 DYNAMIC STRENGTHS. ***
- THE 150 EMP BAR-SPACING RULE DOES NOT APPLY HERE. IT IS A REQUIREMENT OF THE PROTECTIVE ENVELOPE, AND THIS STRUCTURE IS OUTSIDE IT; SPACING IS 200.
- THE OPENING CORNER AT THE TOP OF THE FLIGHT IS DETAILED TO SP 34:1987 Cl. 5.5 - MAIN BARS ARE NOT BENT ROUND IT. SEE DETAIL 3 AND THE CONSTRUCTION REQUIREMENTS PANEL.
- MOVEMENT JOINT WHERE THE STEPPED RAFT MEETS THE HEADHOUSE. THE RAFT BEARS ON COMPACTED FILL AND THE HEADHOUSE BEARS ON THE PRESSURE SLAB - THEY MUST NOT BE CAST MONOLITHIC.
- CONTRACTOR TO CHECK AND VERIFY THE DRG BEFORE EXECUTION.

STRUCTURAL REINFORCEMENT DETAILING - COVERED ENTRY STAIRWELL

CBRN HARDENED UG OPS ROOM

SHEET 15

Project number 01

Date 18 SEP 2026

Drawn by SYN 01

Checked by DR IR CHAUDHARI

STR015

Scale As indicated